

MATH8009 – Regression Tutorial & Lab Exercise Sheet 1

- 1) The following data was taken from claims drawn on life assurance policies for a particular category of employment. It relates the age at official retirement to the age at death for 9 males.

Age of Retirement (x)	57	62	60	57	65	60	58	62	56
Age of Death (y)	71	70	66	70	69	67	69	63	70

$$\sum x = 537, \sum y = 615, \sum xy = 36670, \sum x^2 = 32111, \sum y^2 = 42077$$

- (a) Draw a scatter plot for the above data. Comment on the relationship between the data.
- (b) Calculate the correlation coefficient and comment on your result.
- (c) Calculate the coefficient of determination (or R-squared) and comment on your result.
- (d) Calculate the regression line of y on x . Plot the regression line on the scatter plot from part (a).
- (e) Interpret the slope from part (d).
- (f) Estimate the age of death for a retirement age of
 - (i) 75 years
 - (ii) 59 years
- (g) Which prediction from part (f) do you think is more accurate and why?
- (h) Use R to complete the following:
 - (i) Draw a scatter plot of the data above. Use appropriate labels on the x and y axes. Include the appropriate units of measurement.
 - (ii) Use the `lm` function to confirm the regression line from (d) above.
 - (iii) Use the `abline` function to plot the regression line on the scatter plot.
 - (iv) Use the `cor` function to calculate the correlation coefficient.
 - (v) Calculate the coefficient of determination (or R-squared).
 - (vi) Calculate the mean and median for both the age of retirement and the age of death.

(vii) Plot histograms, one above the other, for the age of retirement and age of death. Use appropriate titles and axes in both histograms. Comment on the shape of the histograms.

(viii) Plot boxplots side by side for the age of retirement and the age of death. From the boxplots read off the values of the first and third quartiles. Comment on the shape of the boxplots.

2) (a) Import the auto_sales.csv dataset (on Canvas) into R; this data contains six variables:

- i. sales (dependent variable y)
- ii. outlets
- iii. reg_auto
- iv. Per_ncome
- v. Av_age_auto
- vi. Supervisors

(b) Use the summary function to find the mean and median for each variable.

(c) Create scatter plots showing the relationship of each variable on the sales. Comment on the relationship between sales and each of the variables.

(d) Use the lm function to create linear models in each case of the above. Write down the regression line for each of the models.

(e) Plot the regression lines from (d) on the appropriate scatter plots.

(f) Use the summary function to determine which models are insignificant? Which model is best and why?

(g) [Advanced] Create a model with sales as the dependent variable which includes all other variables as independent predictors. Interpret this model.

3) As part of an investigation into the level of salaries in her company, a manager decides to tabulate the number of years' service a group of employees have in the company and compare these to the individual employee's salary. The following data was collected.

Years of Service, x	6	4	16	12	10	4	2	18	12
Salary (€000's), y	45	40	90	70	80	45	30	90	50

$$\sum x = 84, \sum y = 540, \sum xy = 5970, \sum x^2 = 1040, \sum y^2 = 36,550$$

(a) Draw a scatter plot for the data in the table.

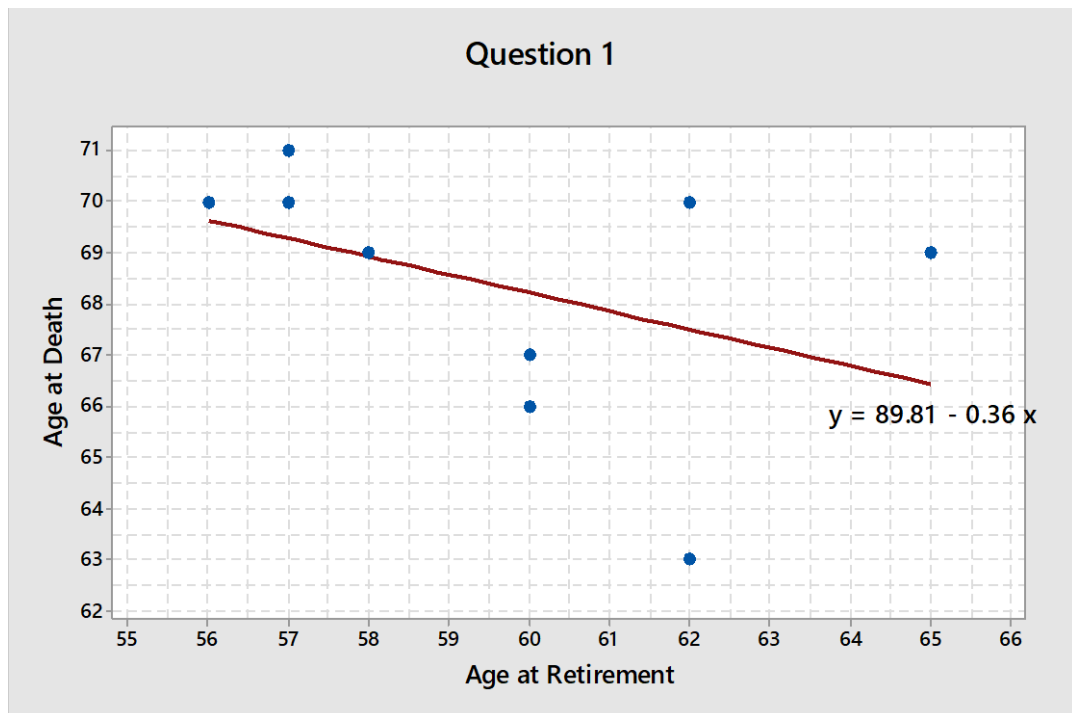
(b) Calculate the least squares line of regression of y on x . Plot the regression line on the scatter diagram from part (a).

(c) Calculate the correlation coefficient and comment on your result.

- (d) If an employee has 11 years of service, estimate a value for their salary using the regression line calculated in part (b).
- (e) Repeat Q1(h) using R for this question.

Solutions

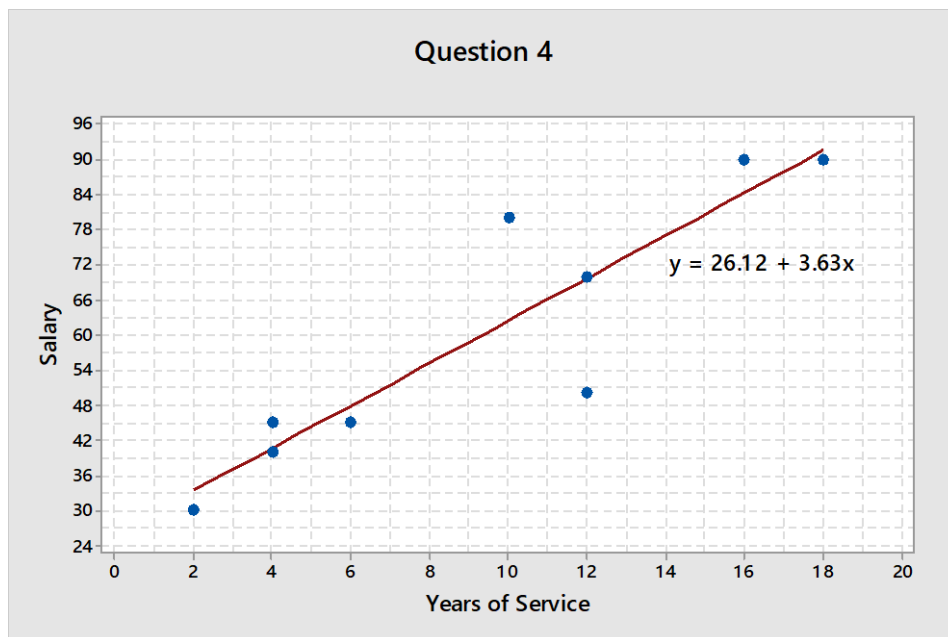
1) (a)



- (b) $r = -0.41$ There is a (slightly weak) negative relationship between the age at retirement and the age at death.
- (c) $R - squared = 0.168$. 16.8% of the variance in the age of death variable can be explained by the retirement age variable.
- (d) $y = 89.81 - 0.36x$
- (e) For every one-year increase in the age of retirement, there is a 0.36 decrease in the age of death.
- (f) (i) 62.81 years (ii) 68.57 years
- (g) Part (ii) is more accurate as it uses interpolation.

2) See R code on Canvas.

3) (a)



(b) $y = 26.12 + 3.63x$

(c) $r = 0.90$ There is a strong positive relationship between an employee's years of service and their salary.

(d) 66.05 (€000's) = €66,050